



SOC ALERT MONITORING AND LOG CORRELATION

Capstone Project Report



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1. Abstract

Security Operations Centers (SOCs) play a vital role in protecting organizational infrastructure by continuously monitoring, detecting, investigating, and responding to cybersecurity threats. As modern environments generate large volumes of authentication, endpoint, and network logs, effective log management and event correlation have become essential for identifying suspicious activities and reducing incident response time. This project focuses on the design and implementation of a SOC monitoring environment using Splunk Enterprise and Sysmon to collect, analyze, and correlate security events from multiple log sources.

A controlled laboratory environment was developed using Windows 10, Sysmon, Splunk Enterprise, Splunk Universal Forwarder, and Kali Linux. Security events were generated through simulated attacker activities to demonstrate the complete SOC monitoring lifecycle. Authentication logs, endpoint telemetry, and network activity logs were collected and analyzed within Splunk. Custom detection rules were developed to generate security alerts for suspicious activities, including Failed Login Detection, System Information Enumeration Detection, and PowerShell Network Reconnaissance Detection.

The generated alerts were investigated and correlated based on event timestamps and activity patterns to reconstruct attacker behavior and establish an incident investigation workflow. This correlation process enabled the identification of relationships between multiple security events and demonstrated how SOC analysts perform threat detection, triage, and incident analysis. In addition, dashboards were created to provide centralized visibility into security events and alert activity.

As a future enhancement, an AI-powered incident reporting workflow was implemented using n8n and Google Gemini. The solution automatically collects correlated alerts, analyzes incident data, generates structured incident reports, and delivers them through email, reducing manual reporting effort and improving operational efficiency.

The project successfully demonstrates the practical implementation of SOC monitoring, log collection, alert generation, event correlation, investigation workflows, and AI-assisted incident reporting. The results highlight the importance of centralized security monitoring and demonstrate how automation and artificial intelligence can enhance modern SOC operations.

2. Introduction

A Security Operations Center (SOC) is a centralized unit responsible for continuously monitoring, detecting, analyzing, and responding to cybersecurity incidents within an organization. SOC analysts monitor security events generated from various sources such as endpoints, servers, network devices, and applications to identify potential threats and suspicious activities.

The primary objective of a SOC is to improve an organization's security posture by providing real-time visibility into security events and enabling rapid incident response. SOC teams use specialized tools such as Security Information and Event Management (SIEM) platforms to collect and analyze logs from multiple systems.

Security Information and Event Management (SIEM) is a cybersecurity solution that combines Security Information Management (SIM) and Security Event Management (SEM) capabilities. SIEM platforms collect, store, analyze, and correlate logs from various devices and applications across an organization's infrastructure.

SIEM solutions help security analysts identify suspicious activities, investigate incidents, and generate alerts based on predefined correlation rules. Popular SIEM platforms include Splunk, IBM QRadar, Microsoft Sentinel, ArcSight, and Elastic SIEM. In this project, Splunk Enterprise was used as the primary SIEM platform for log collection, monitoring, and alert generation.

Modern organizations generate thousands of security events every day. Individual security alerts often provide limited information and may not accurately represent the complete attack scenario. Log correlation is the process of linking multiple related events from different log sources to identify patterns and reconstruct attacker activities.

For example, a failed login attempt followed by system reconnaissance and network scanning may indicate a potential attack sequence. By correlating these events based on timestamps and system activity, security analysts can gain better visibility into the incident and perform more effective investigations.

This project focuses on developing a SOC monitoring environment using Splunk Enterprise and Sysmon. Authentication logs, endpoint logs, and network logs were collected and analyzed to generate security alerts. Multiple events were correlated to reconstruct attacker behavior and demonstrate the investigation workflow followed by SOC analysts.

Additionally, an AI-powered incident reporting workflow was implemented as a future enhancement using n8n and Google Gemini. The workflow automatically analyzes correlated alerts and generates structured incident reports, reducing manual effort and improving incident response efficiency.

3. Objectives

The primary objective of this project is to design and implement a Security Operations Center (SOC) monitoring environment capable of collecting, analyzing, and correlating security events from multiple log sources. The project aims to demonstrate the practical application of Security Information and Event Management (SIEM) technologies in detecting suspicious activities, investigating security incidents, and improving organizational security visibility through centralized monitoring.

With the increasing volume and complexity of cybersecurity threats, organizations require efficient mechanisms to monitor systems, identify malicious activities, and respond to incidents in a timely manner. This project focuses on building a SOC-oriented monitoring solution using Splunk Enterprise and Sysmon to simulate real-world security monitoring and incident investigation processes. The implementation provides hands-on experience in log management, alert generation, event correlation, and incident analysis while demonstrating the responsibilities performed by SOC analysts in a modern security environment.

The specific objectives of this project are as follows:

- To design and deploy a SOC monitoring environment using Splunk Enterprise as the SIEM platform and Sysmon for endpoint activity monitoring.
- To collect, centralize, and analyze authentication, endpoint, and network security logs generated from multiple sources within the laboratory environment.
- To create and configure detection rules capable of identifying suspicious activities and generating real-time security alerts.
- To simulate attacker-like activities in a controlled environment and evaluate the effectiveness of the monitoring and detection mechanisms.
- To correlate multiple security events and reconstruct attack sequences in order to understand attacker behavior and improve investigation accuracy.
- To demonstrate the incident triage and investigation workflow commonly followed by Security Operations Center analysts.
- To develop dashboards that provide centralized visibility into security events, alerts, and system activities for effective monitoring and analysis.
- To evaluate the impact of log correlation on incident detection, investigation efficiency, and overall security visibility.
- To explore the integration of automation and artificial intelligence by implementing an AI-assisted incident reporting workflow using n8n and Google Gemini as a future enhancement

4. Tools And Technologies

This project utilizes a combination of security monitoring, log management, endpoint monitoring, and automation tools to establish a functional Security Operations Center (SOC) environment. Each tool plays a specific role in the collection, analysis, monitoring, and investigation of security events.

4.1 Splunk Enterprise

Splunk Enterprise was used as the primary Security Information and Event Management (SIEM) platform for this project. It provides centralized log collection, storage, analysis, and visualization capabilities, enabling security analysts to monitor and investigate security events across multiple systems.

Purpose:

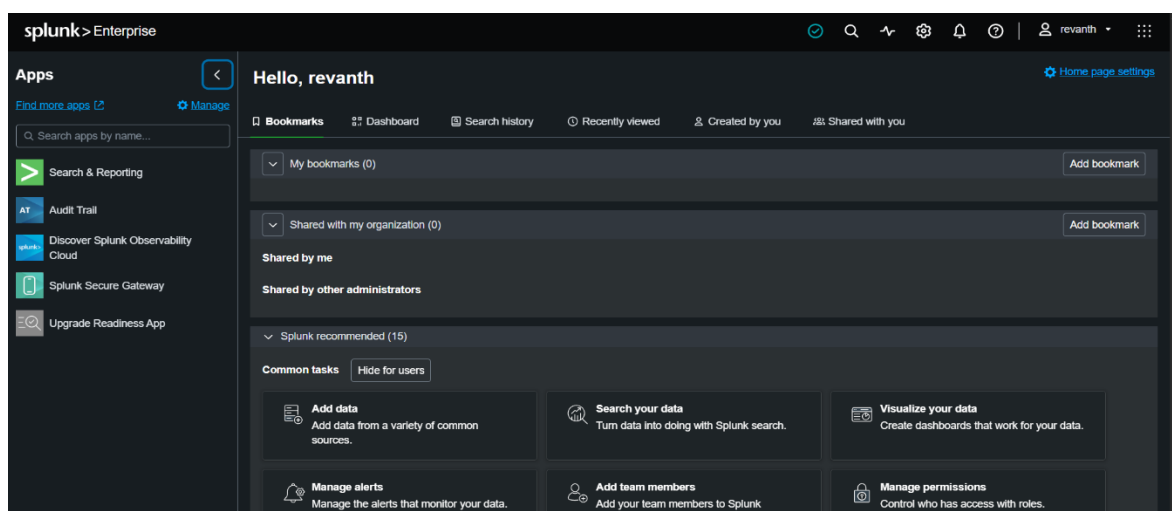
To collect, index, search, analyze, and correlate security logs from various sources.

Key Features:

- Centralized log collection and management
- Real-time alert generation
- Dashboard creation and visualization
- Event correlation and investigation
- Powerful search and reporting capabilities

Role in the Project:

Splunk Enterprise received logs from the Windows endpoint through the Splunk Universal Forwarder. The platform was used to create detection rules, generate security alerts, perform log correlation, and develop dashboards for monitoring suspicious activities.



4.2 Sysmon

Sysmon (System Monitor) is a Microsoft Sysinternals tool that provides detailed visibility into endpoint activities. It generates advanced Windows event logs that assist security analysts in identifying suspicious processes, network connections, and system modifications.

Purpose:

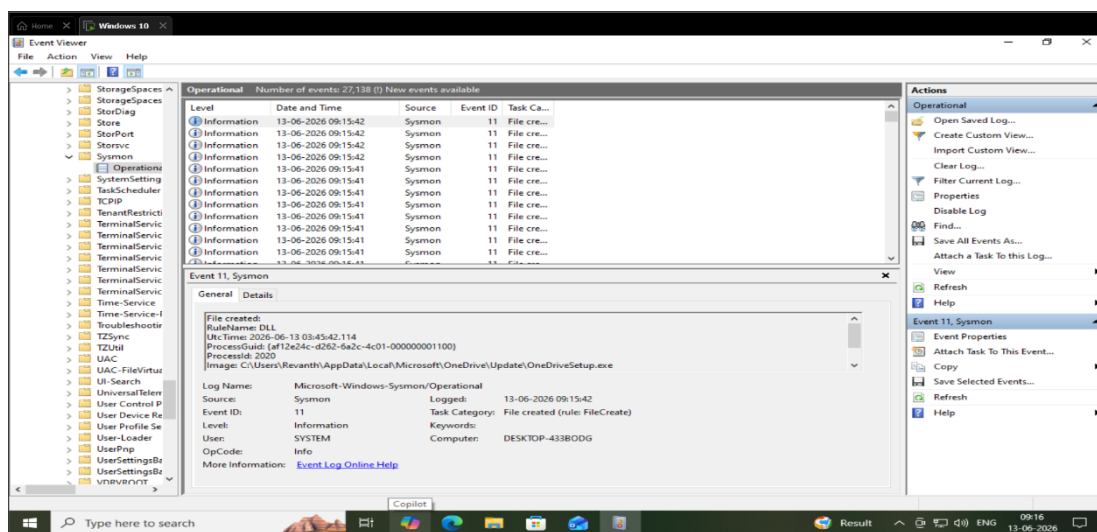
To monitor and record detailed endpoint activities.

Key Features:

- Process creation monitoring
- Network connection monitoring
- Command-line activity logging
- Detailed endpoint telemetry
- Enhanced visibility into system events

Role in the Project:

Sysmon was installed on the Windows 10 endpoint to generate detailed logs for process executions and network activities. These logs were forwarded to Splunk Enterprise for analysis and alert generation.



4.3 Splunk Universal Forwarder

The Splunk Universal Forwarder is a lightweight log collection agent that securely forwards logs from endpoints to a centralized Splunk server.

Purpose:

To collect and forward Windows and Sysmon logs to Splunk Enterprise.

Key Features:

- Lightweight and efficient log forwarding
- Secure transmission of log data
- Real-time log collection
- Support for multiple log sources

Role in the Project:

The Universal Forwarder was installed on the Windows endpoint and configured to forward Windows Security Logs and Sysmon logs to the Splunk Enterprise server for centralized monitoring and analysis.



4.4 Kali Linux

Kali Linux is a security-focused operating system widely used for penetration testing, vulnerability assessment, and cybersecurity research.

Purpose:

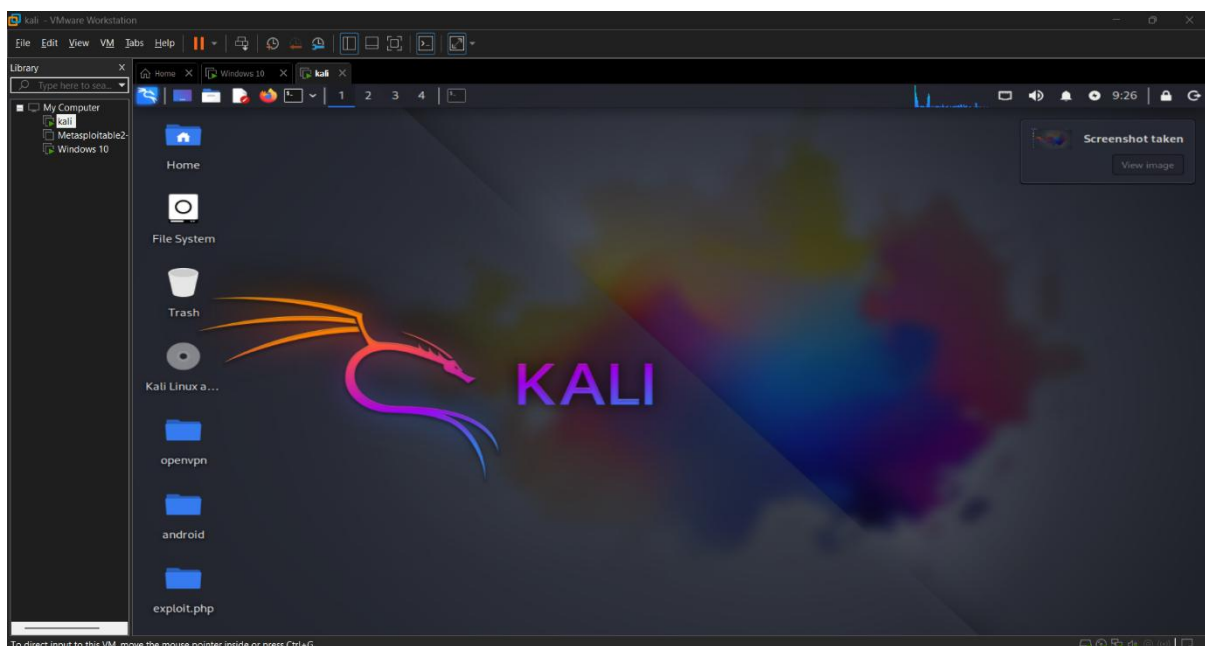
To simulate attacker activities and generate security events within the laboratory environment.

Key Features:

- Security testing and assessment tools
- Network analysis utilities
- Penetration testing framework support
- Traffic generation and attack simulation capabilities

Role in the Project:

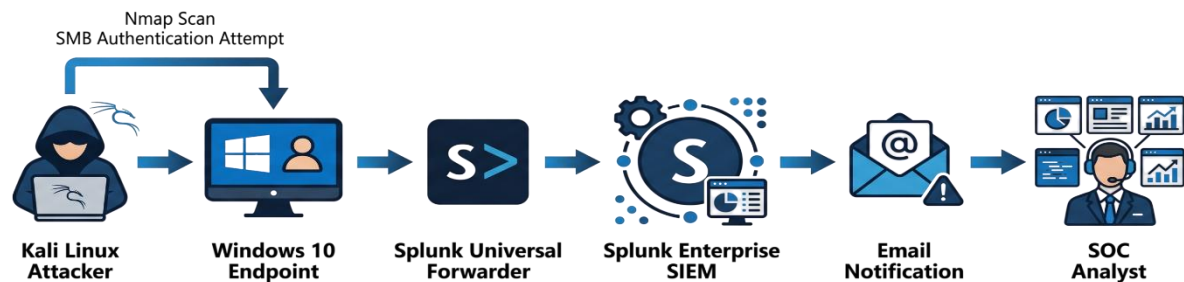
Kali Linux was used as the attack simulation machine to generate controlled security events such as failed authentication attempts and network reconnaissance activities. These activities produced logs that were collected and analyzed by Splunk Enterprise.



5. Lab Setup And Architecture

To simulate a real-world Security Operations Center (SOC) environment, a laboratory setup was developed consisting of an attacker machine, a monitored endpoint, log collection agents, and a centralized SIEM platform. The architecture was designed to demonstrate the complete security monitoring lifecycle, from event generation and log collection to alert generation and incident investigation.

The environment consists of the following components:



Architecture Components

1. Kali Linux

Kali Linux was used as the attack simulation machine to generate controlled security events within the laboratory environment. Various test activities were performed to simulate suspicious behavior and produce security logs that could be detected and analyzed by the monitoring infrastructure.

2. Windows 10 Endpoint

The Windows 10 system served as the monitored endpoint. Security events generated on the system, including authentication activities, process executions, and network connections, were recorded and used for security monitoring and investigation.

3. Sysmon

Sysmon (System Monitor) was deployed on the Windows endpoint to provide detailed visibility into system activities. It generated enhanced telemetry such as process creation events, network connections, and other endpoint activities that are not fully available through standard Windows logging.

4. Splunk Universal Forwarder

The Splunk Universal Forwarder was installed on the Windows endpoint to securely collect and forward logs to the Splunk Enterprise server. This ensured centralized log aggregation and enabled real-time monitoring of security events.

5. Splunk Enterprise

Splunk Enterprise served as the Security Information and Event Management (SIEM) platform. It collected logs from multiple sources, indexed security events, generated alerts based on predefined detection rules, and provided dashboards for monitoring and

investigation. Splunk was also used to correlate multiple events and reconstruct attacker behavior during the investigation process.

Working of the Architecture

The monitoring workflow begins when activities occur on the Windows endpoint. These activities generate security logs that are captured by Sysmon and Windows Event Logs. The Splunk Universal Forwarder collects these logs and sends them to Splunk Enterprise for analysis. Splunk processes the incoming data, applies detection rules, generates security alerts, and enables analysts to investigate suspicious activities through dashboards and correlation searches.

This architecture provides a centralized monitoring environment capable of detecting suspicious behavior, generating actionable alerts, and supporting incident investigation through log correlation and security analysis.

6. Data Collection

6.1 Authentication Logs

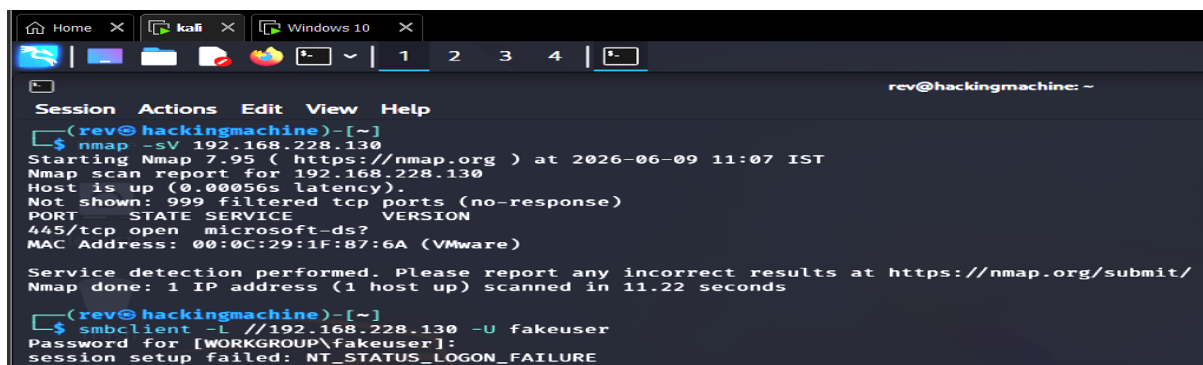
Authentication logs were collected from the Windows Security Log to monitor user login activities and identify unauthorized access attempts. These logs provide visibility into authentication-related events occurring on the endpoint and are commonly used by SOC analysts to detect credential-based attacks.

Source: Windows Security Log

Event ID: 4625

Purpose: Detect failed login attempts and potential brute-force or unauthorized access activities.

Analysis: Event ID 4625 is generated whenever a login attempt fails. Monitoring these events helps security analysts identify suspicious authentication activity, password guessing attacks, and potential account compromise attempts.



```
rev@hackingmachine: ~  
$ nmap -sV 192.168.228.130  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-09 11:07 IST  
Nmap scan report for 192.168.228.130  
Host is up (0.00056s latency).  
Not shown: 999 filtered tcp ports (no-response)  
PORT      STATE SERVICE      VERSION  
445/tcp    open  microsoft-ds?    
MAC Address: 00:0C:29:1F:87:6A (VMware)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/  
Nmap done: 1 IP address (1 host up) scanned in 11.22 seconds  
  
$ smbclient -L //192.168.228.130 -U fakeuser  
Password for [WORKGROUP\fakeuser]:  
session setup failed: NT_STATUS_LOGON_FAILURE
```

i	Time	Event
>	6/9/26 11:11:30.647 AM	06/09/2026 11:11:30.647 AM LogName=Security EventCode=4625 EventType=0 ComputerName=DESKTOP-433BODG Show all 61 lines host = DESKTOP-433BODG source = WinEventLog:Security sourcetype = WinEventLog:Security

6.2 Endpoint Logs

Endpoint activity was monitored using Sysmon to provide detailed visibility into process execution and system-level events. These logs help analysts identify suspicious commands and reconnaissance activities performed on the host.

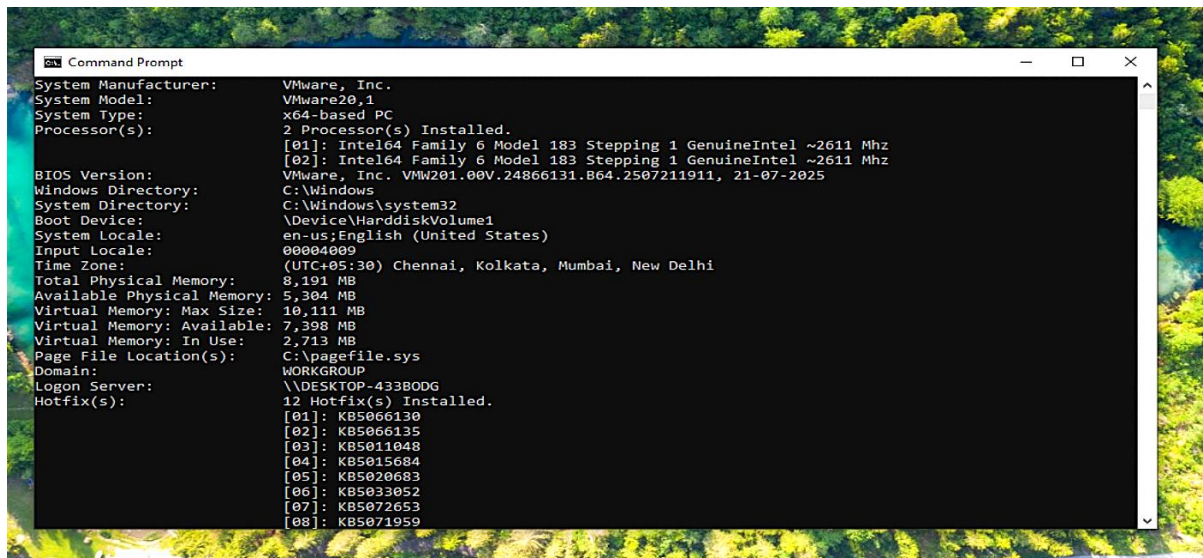
Source: Sysmon

Event ID: 1

Process: systeminfo.exe

Purpose: Detect system information discovery and host reconnaissance activities.

Analysis: The execution of the systeminfo.exe command may indicate that a user or attacker is gathering information about the operating system, hostname, architecture, installed updates, and system configuration. Such activities are commonly observed during the reconnaissance phase of an attack.



```

Command Prompt
System Manufacturer: VMware, Inc.
System Model: VMware20,1
System Type: x64-based PC
Processor(s): 2 Processor(s) Installed.
               [01]: Intel64 Family 6 Model 183 Stepping 1 GenuineIntel ~2611 Mhz
               [02]: Intel64 Family 6 Model 183 Stepping 1 GenuineIntel ~2611 Mhz
BIOS Version: VMware, Inc. VMW201.00V.24866131.864.2507211911, 21-07-2025
Windows Directory: C:\Windows
System Directory: C:\Windows\system32
Boot Device: \Device\HarddiskVolume1
System Locale: en-us;English (United States)
Input Locale: 00004009
Time Zone: (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory: 8,191 MB
Available Physical Memory: 5,304 MB
Virtual Memory: Max Size: 10,111 MB
Virtual Memory: Available: 7,398 MB
Virtual Memory: In Use: 2,713 MB
Page File Location(s): C:\pagefile.sys
Domain: WORKGROUP
Logon Server: \\DESKTOP-433B0DG
Hotfix(s): 12 Hotfix(s) Installed.
            [01]: KB5066130
            [02]: KB5066135
            [03]: KB5011048
            [04]: KB5015684
            [05]: KB5020683
            [06]: KB5033052
            [07]: KB5072653
            [08]: KB5071959
  
```

i	Time	Event
>	6/9/26 11:17:01.568 AM	06/09/2026 11:17:01.568 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=1 EventType=4 ComputerName=DESKTOP-433B0DG Show all 38 lines host = DESKTOP-433B0DG source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational

6.3 Network Logs

Network-related activities were monitored using Sysmon network connection events. These logs provide visibility into outbound and inbound network communications initiated by processes running on the endpoint.

Source: Sysmon

Event ID: 3

Process: PowerShell.exe

Purpose: Detect PowerShell-based network reconnaissance and suspicious network connections.

Analysis: Event ID 3 records network connections initiated by processes. Monitoring PowerShell network activity helps identify reconnaissance, service discovery, remote communication attempts, and other suspicious behaviors that may indicate attacker activity.

```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\Revanth> Test-NetConnection 192.168.228.128 -Port 8080

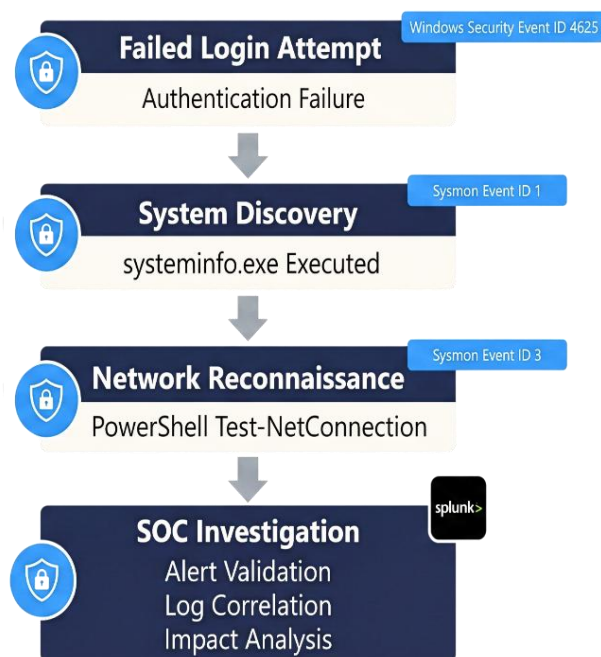
ComputerName      : 192.168.228.128
RemoteAddress     : 192.168.228.128
RemotePort        : 8080
InterfaceAlias    : Ethernet0
SourceAddress     : 192.168.228.130
TcpTestSucceeded  : True

PS C:\Users\Revanth>

```

i	Time	Event
>	6/9/26 11:24:19:155 AM	06/09/2026 11:24:19.155 AM LogName=Microsoft-Windows-Sysmon/Operational EventCode=3 EventType=4 ComputerName=DESKTOP-433BODG Show all 33 lines host = DESKTOP-433BODG source = WinEventLog:Microsoft-Windows-Sysmon/Operational sourcetype = WinEventLog:Microsoft-Windows-Sysmon/Operational

Workflow :



7. Alert Generation

To validate the effectiveness of the SOC monitoring environment, multiple controlled security activities were performed within the laboratory environment. These activities were designed to generate security events that could be collected, analyzed, and detected by Splunk Enterprise. Alert rules were configured to identify suspicious behavior and notify analysts when predefined conditions were met.

7.1 Failed Login Detection

The first alert was designed to detect failed authentication attempts that may indicate unauthorized access attempts, password guessing, or brute-force activities.

Objective:

To monitor failed login attempts and identify suspicious authentication activity.

Activity Performed:

An invalid authentication attempt was generated against the Windows endpoint from the testing environment. This activity caused Windows to record a failed login event in the Security Log.

Event Generated:

- Source: Windows Security Log
- Event ID: 4625
- Description: Failed Login Attempt

Detection Logic:

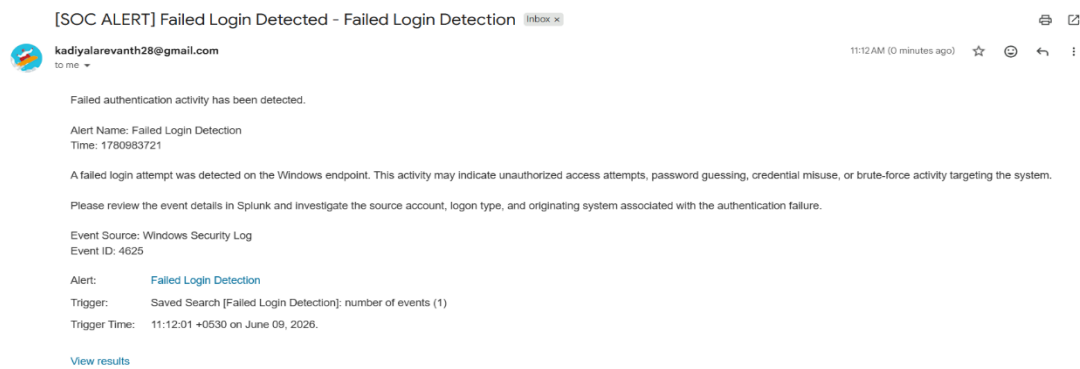
Splunk continuously monitored Windows Security Logs and generated an alert whenever Event ID 4625 was detected.

Alert Generated:

Failed Login Detection

Analysis:

Failed authentication attempts are often one of the earliest indicators of unauthorized access attempts. Monitoring these events helps SOC analysts identify potential credential attacks and investigate suspicious login activity before a compromise occurs.



7.2 System Information Enumeration

The second alert focused on detecting system reconnaissance activities commonly performed after gaining access to a system. Attackers often gather information about the operating system, hostname, architecture, and installed updates before proceeding with further actions.

Objective:

To identify system discovery and reconnaissance activities performed on the endpoint.

Activity Performed:

A system information query was executed on the Windows endpoint to collect operating system and hardware details.

Event Generated:

- Source: Sysmon
- Event ID: 1
- Process: systeminfo.exe

Detection Logic:

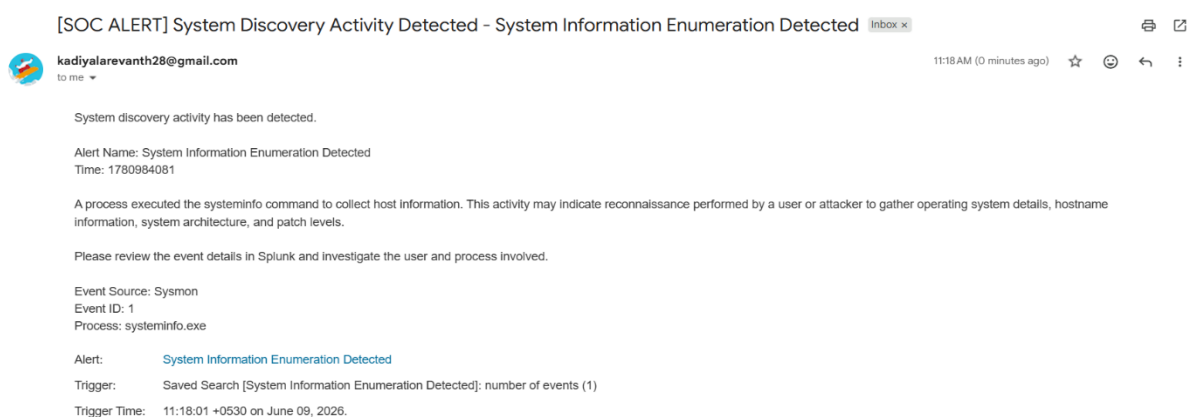
Splunk monitored Sysmon process creation events and generated an alert whenever the systeminfo.exe process was executed.

Alert Generated:

System Information Enumeration Detected

Analysis:

The execution of systeminfo.exe may indicate information-gathering activity. Monitoring such commands helps analysts detect reconnaissance techniques commonly associated with the early stages of an attack.



7.3 PowerShell Network Reconnaissance

The third alert was designed to detect network reconnaissance activities performed using PowerShell. Network discovery is commonly used to identify reachable systems, services, and communication paths within an environment.

Objective:

To detect suspicious network connections initiated through PowerShell.

Activity Performed:

A PowerShell-based network connectivity test was performed, generating a network connection event that could be monitored through Sysmon.

Event Generated:

- Source: Sysmon
- Event ID: 3
- Process: PowerShell.exe

Detection Logic:

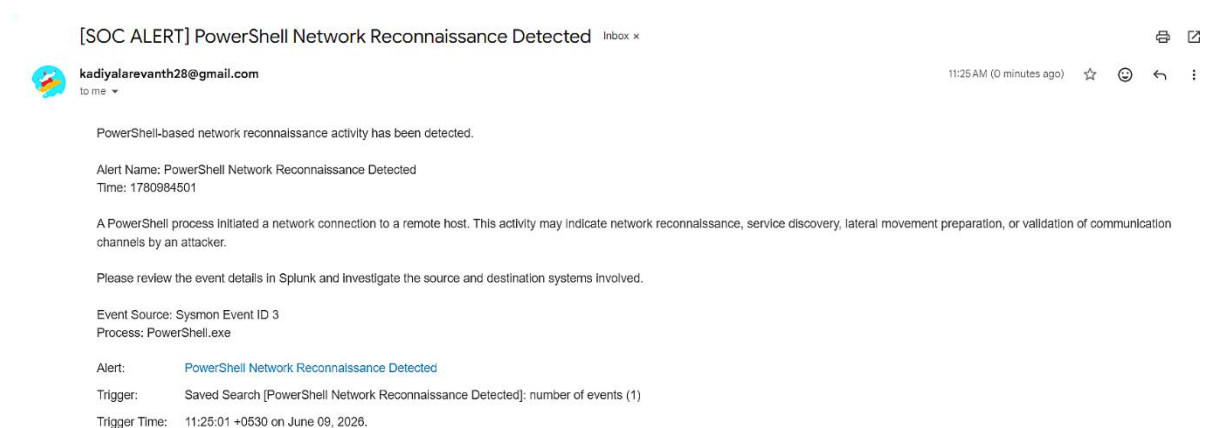
Splunk monitored Sysmon network connection events and generated an alert whenever PowerShell initiated a network connection matching the configured detection criteria.

Alert Generated:

PowerShell Network Reconnaissance Detected

Analysis:

PowerShell is a legitimate administrative tool but is also frequently used during security incidents for reconnaissance and network discovery. Monitoring PowerShell network activity helps identify potentially suspicious behavior and improves visibility into endpoint communications.



8. Log Correlation And Investigation

After generating the security alerts, a correlation analysis was performed to determine whether the events were related and to understand the sequence of activities occurring on the monitored endpoint. Log correlation is a critical SOC function that helps analysts connect multiple security events and reconstruct potential attacker behavior.

8.1 Alert Correlation Timeline

The generated alerts were reviewed and arranged chronologically based on their timestamps.

15:11 – Failed Login Detection (Windows Event ID 4625)



15:17 – System Information Enumeration Detected (Sysmon Event ID 1)



15:24 – PowerShell Network Reconnaissance Detected (Sysmon Event ID 3)

The screenshot shows the Splunk Enterprise interface for a 'SOC Log Correlation Investigation'. The search bar contains a query for EventCode=4625, OR (EventCode=1 Image=*systeminfo.exe), OR (EventCode=3 Image=*powershell.exe). The results show 3 events from 6/9/26 10:59:19.000 AM to 6/9/26 11:28:19.000 AM. The events are listed in a table with columns: _time, Category, EventCode, Image, User, and ComputerName.

_time	Category	EventCode	Image	User	ComputerName
2026-06-09 11:11:38.647	Authentication	4625		NOT_TRANSLATED	DESKTOP-433B00G
2026-06-09 11:17:01.568	Endpoint	1	C:\Windows\System32\systeminfo.exe	DESKTOP-433B00G\Revanth	DESKTOP-433B00G
2026-06-09 11:24:19.155	Network	3	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	DESKTOP-433B00G\Revanth	DESKTOP-433B00G

The timeline shows a progression from authentication-related activity to endpoint reconnaissance and finally network reconnaissance. While each alert individually may represent legitimate activity, correlating multiple alerts provides greater visibility into potentially suspicious behavior.

8.2 Investigation Methodology

A structured investigation process was followed to analyze the alerts and determine their relationship.

Step 1: Authentication Log Review

The investigation began by examining Windows Security Logs to identify failed authentication attempts. Event ID 4625 was analyzed to determine the source and timing of the failed login activity.

Objective:

Identify potential unauthorized access attempts and suspicious authentication behavior.

Evidence Collected:

Windows Security Event ID 4625

Step 2: Process Execution Analysis

Sysmon process creation logs were reviewed to identify commands executed on the endpoint following the failed login activity. The execution of systeminfo.exe was detected and investigated.

Objective:

Determine whether system reconnaissance activity occurred on the endpoint.

Evidence Collected:

Sysmon Event ID 1

Process: systeminfo.exe

Step 3: Network Activity Analysis

Network connection logs generated by Sysmon were reviewed to identify outbound communication initiated by PowerShell.

Objective:

Identify potential network reconnaissance or suspicious communications.

Evidence Collected:

Sysmon Event ID 3

Process: PowerShell.exe

Step 4: Timestamp Correlation

The timestamps of all alerts were compared to determine whether the events occurred within a related timeframe.

Observation:

- Failed Login detected at 15:11
- System Information Enumeration detected at 15:17
- PowerShell Network Reconnaissance detected at 15:24

The events occurred within a short time window, indicating a possible sequence of related activities.

_time	Category	EventCode	Image	User	ComputerName
2026-06-09 11:11:30.647	Authentication	4625			DESKTOP-433B00G
2026-06-09 11:17:01.568	Endpoint	1	C:\Windows\System32\systeminfo.exe	NOT_TRANSLATED DESKTOP-433B00G\Revanth	DESKTOP-433B00G
2026-06-09 11:24:19.155	Network	3	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	NOT_TRANSLATED DESKTOP-433B00G\Revanth	DESKTOP-433B00G

Step 5: Investigation Timeline Construction

Based on the collected evidence, an investigation timeline was created to visualize the progression of events.

Correlated Sequence:

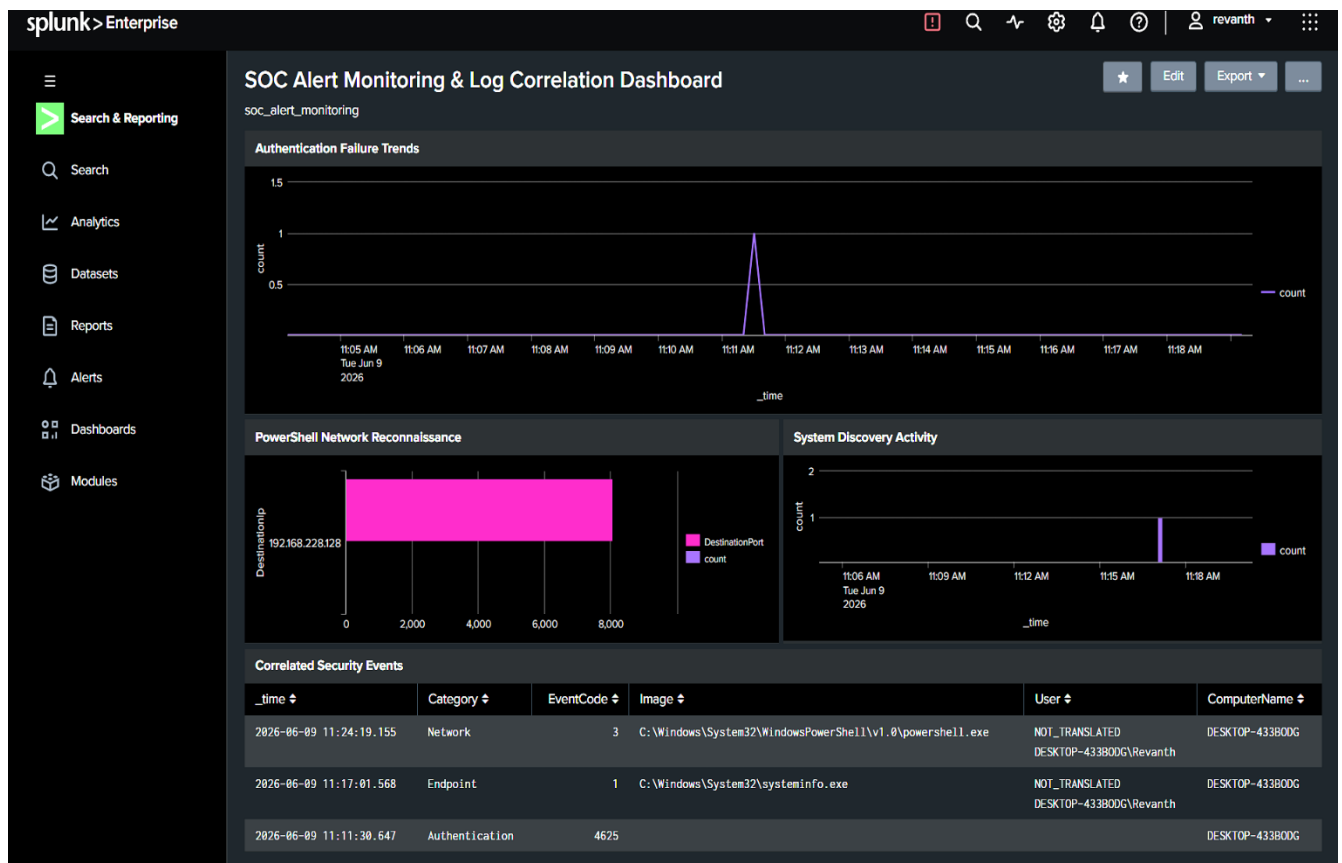
1. Failed authentication attempt observed.

2. System information gathering activity detected.
3. PowerShell network communication activity identified.
4. Events correlated using timestamps and log analysis.
5. Investigation report generated for analyst review.

9. Dashboard Monitoring

Security dashboards play a vital role in Security Operations Centers (SOCs) by providing real-time visibility into security events, alerts, and system activities. A centralized dashboard enables security analysts to quickly identify suspicious behavior, monitor trends, and prioritize incident investigations.

In this project, Splunk dashboards were developed to provide a consolidated view of authentication events, endpoint activities, network events, and generated security alerts. The dashboards assisted in monitoring the overall security posture of the laboratory environment and supported the investigation process.



9.1 Authentication Monitoring

The Authentication Monitoring dashboard was created to track user login activities and identify failed authentication attempts.

Purpose:

To monitor authentication-related events and detect suspicious login activity.

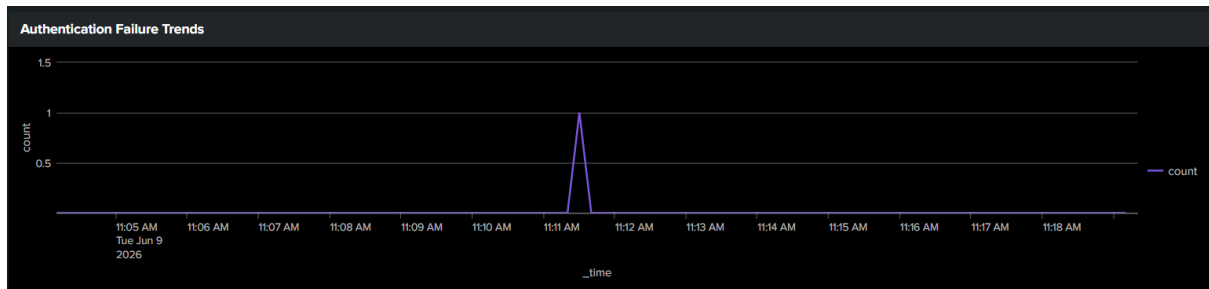
Monitored Events:

- Successful login events
- Failed login attempts
- Authentication trends

- Security Event ID 4625

Benefits:

- Early detection of unauthorized access attempts
- Identification of brute-force activities
- Improved visibility into user authentication behavior



9.2 Endpoint Monitoring

The Endpoint Monitoring dashboard was designed to provide visibility into process execution and host-level activities collected through Sysmon.

Purpose:

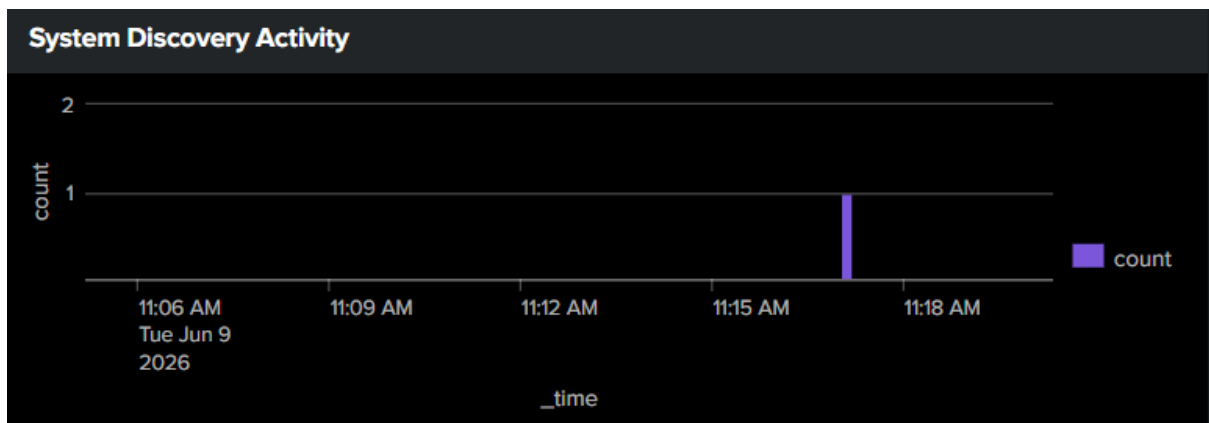
To monitor endpoint activities and detect suspicious process executions.

Monitored Events:

- Process creation events
- Command-line executions
- System discovery activities
- Sysmon Event ID 1

Benefits:

- Detection of reconnaissance activities
- Improved endpoint visibility
- Identification of suspicious processes



9.3 Network Monitoring

The Network Monitoring dashboard provided visibility into network connections initiated by processes running on the endpoint.

Purpose:

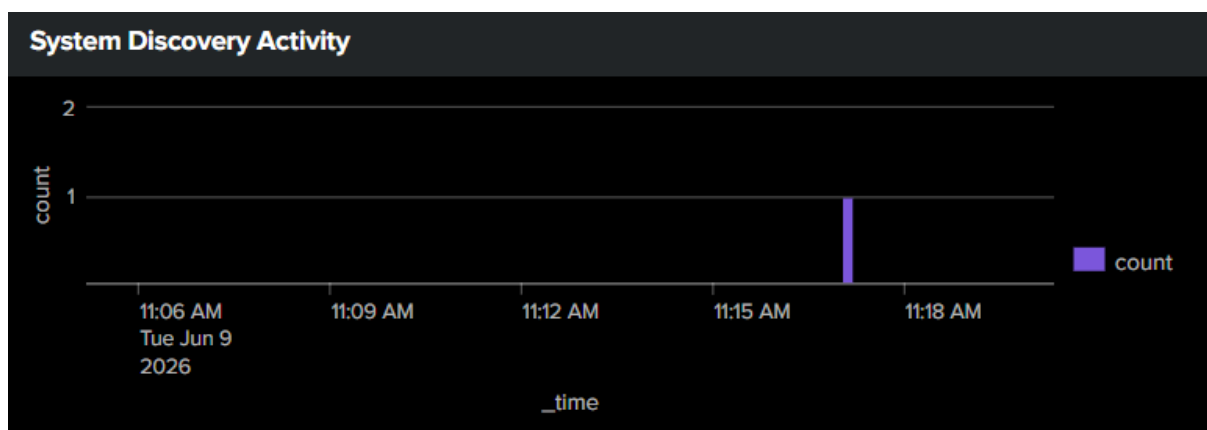
To monitor network communications and identify suspicious connections.

Monitored Events:

- Outbound network connections
- PowerShell network activity
- Sysmon Event ID 3
- Network reconnaissance activities

Benefits:

- Detection of suspicious communications
- Identification of network scanning behavior
- Enhanced network visibility



9.4 Alert Statistics Dashboard

The Alert Statistics dashboard provided a consolidated view of all generated security alerts and their frequency.

Purpose:

To monitor alert trends and support incident prioritization.

Displayed Information:

- Total alerts generated
- Alert categories
- Alert frequency
- Security event trends

Benefits:

- Centralized alert management
- Faster incident triage
- Improved security monitoring efficiency

Correlated Security Events					
_time ↕	Category ↕	EventCode ↕	Image ↕	User ↕	ComputerName ↕
2026-06-09 11:24:19.155	Network	3	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	NOT_TRANSLATED DESKTOP-433BODG\Revanth	DESKTOP-433BODG
2026-06-09 11:17:01.568	Endpoint	1	C:\Windows\System32\systeminfo.exe	NOT_TRANSLATED DESKTOP-433BODG\Revanth	DESKTOP-433BODG
2026-06-09 11:11:30.647	Authentication	4625			DESKTOP-433BODG

Dashboard Benefits

The dashboards developed in this project provided centralized monitoring and improved visibility across authentication, endpoint, and network security events. By consolidating multiple log sources into a single interface, analysts were able to identify suspicious activities more efficiently, investigate alerts faster, and perform effective log correlation.

The dashboard environment demonstrated how SIEM platforms support real-time security monitoring and enhance the overall effectiveness of SOC operations.

10. AI-Powered Incident Reporting

Modern Security Operations Centers (SOCs) generate a large number of security alerts every day. Manually reviewing, correlating, and documenting incidents can be time-consuming and may delay the investigation process. To address this challenge, an AI-assisted incident reporting workflow was developed as a future enhancement to automate alert analysis and incident report generation.

The solution integrates Splunk, Gmail, n8n, and Google Gemini to automatically collect related alerts, perform event correlation, analyze security events, and generate structured incident reports. This automation reduces manual effort and helps SOC analysts focus on higher-priority investigation activities.

10.1 AI Workflow Architecture



10.2 Workflow Execution

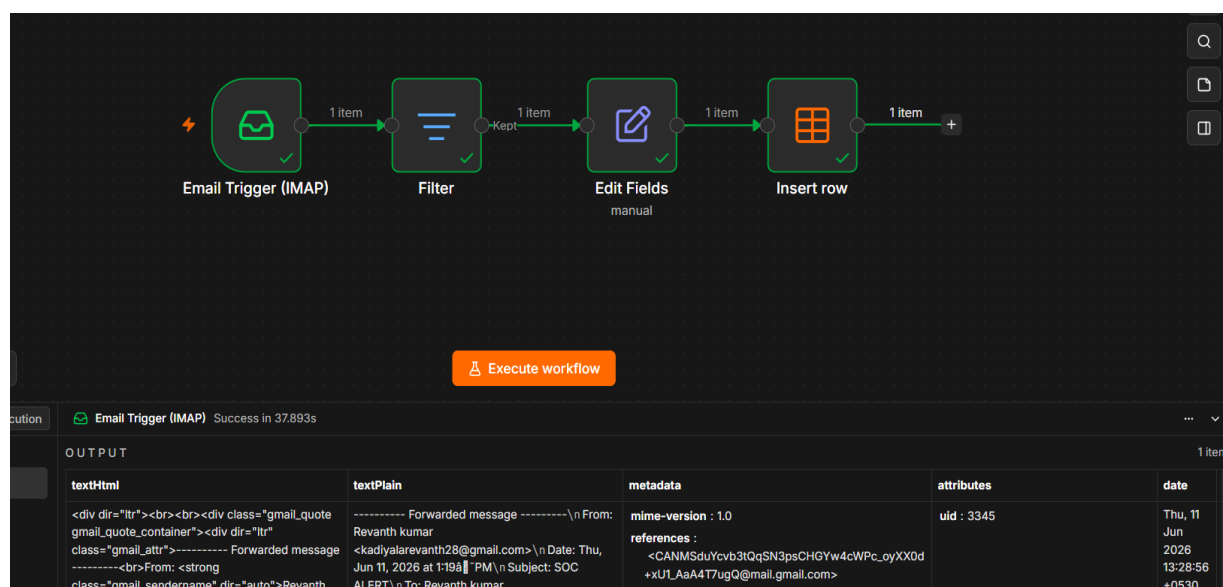
The automated workflow operates as follows:

Step 1: Alert Generation

Splunk continuously monitors incoming logs and generates security alerts whenever suspicious activities match predefined detection rules.

Step 2: Email Notification

Generated alerts are automatically delivered to a dedicated Gmail inbox through Splunk's email alerting mechanism.



Step 3: Alert Collection

The n8n automation platform monitors the mailbox and retrieves newly received alert emails.

Step 4: Alert Storage

Collected alerts are stored within an n8n Data Table to maintain a centralized repository of security events.

Step 5: Event Correlation

n8n groups and correlates related alerts occurring within a predefined time window to identify potentially connected activities.

☐	☆ me	SOC Incident Report - AI Analysis - ### 1. Executive Summary On June 11, 2026, a series of security alerts were triggered involvi...	5:13 PM
☐	☆ me	[SOC ALERT] PowerShell Network Reconnaissance Detected - PowerShell-based network reconnaissance activity has been det...	5:10 PM
☐	☆ me	[SOC ALERT] Failed Login Detected - Failed Login Detection - Failed authentication activity has been detected. Alert Name: Fail...	5:09 PM
☐	☆ me	[SOC ALERT] System Discovery Activity Detected - System Information Enumeration Detected - System discovery activity has b...	5:09 PM

Step 6: AI Analysis

The correlated alert data is sent to Google Gemini for analysis. The AI model reviews the alerts and generates a structured incident assessment based on the provided evidence.

Step 7: Incident Report Generation

Google Gemini produces an incident report containing:

- Executive Summary
- Attack Timeline
- Severity Assessment
- Impact Analysis
- MITRE ATT&CK Mapping
- Recommendations

The screenshot displays the n8n workflow editor for a workflow titled "AI SOC REPORT - EMAIL". The workflow consists of the following steps:

- Schedule Trigger**: Initiates the workflow on a schedule.
- Get row(s)**: Retrieves data from a database table, outputting 3 items.
- Aggregate**: Processes the retrieved data, outputting 1 item.
- Message a model**: Sends the aggregated data to an AI model (Google Gemini) for analysis. The message is of type "text".
- Send a message**: Sends the output from the AI model via email.

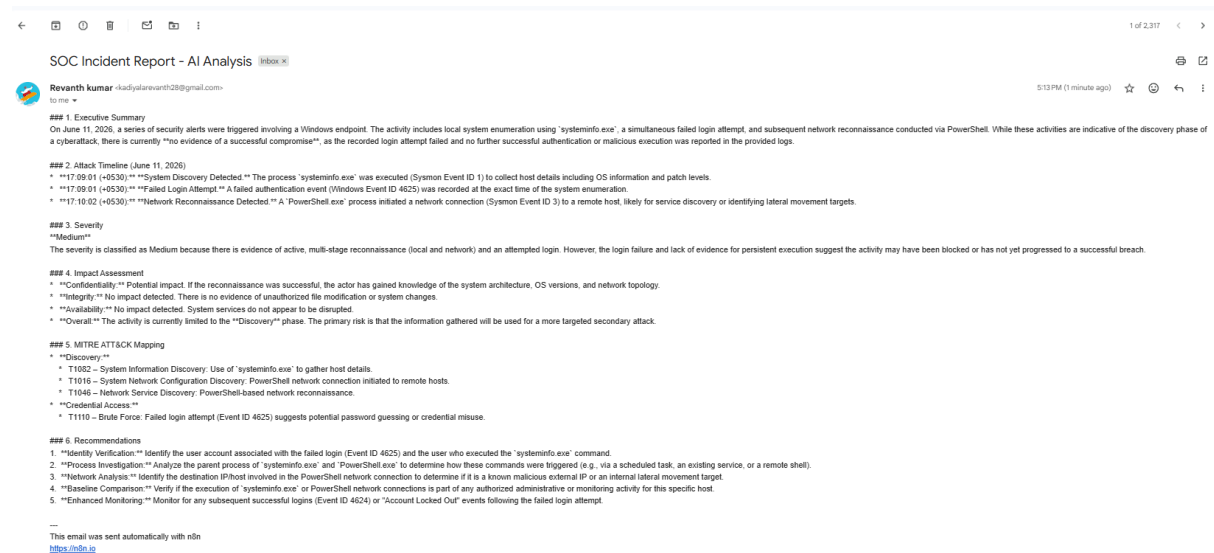
The "Message a model" step is currently selected, and its output is visible in the "OUTPUT" pane. The output is a structured JSON object:

```
{  "content": "text: ### 1. Executive Summary\\nOn June 11, 2026, a series of security alerts were triggered involving a series of security alerts were triggered involving a Windows endpoint. The activity includes local system enumeration using 'systeminfo.exe', a",  "finishReason": "STOP",  "index": 0}
```

The "Log" pane shows the execution history, indicating that the workflow was successfully executed on June 11, 2026, at 17:14.

Step 8: Report Delivery

The generated incident report is automatically sent via email to the SOC analyst for review and further investigation



10.3 Benefits of AI-Assisted Reporting

- Faster incident triage and analysis
- Automated alert correlation
- Reduced manual reporting effort
- Consistent incident documentation
- Improved investigation efficiency
- Enhanced SOC productivity

10.4 Future Scope

Future improvements may include automated ticket creation, SOAR integration, threat intelligence enrichment, and AI-assisted remediation recommendations to further enhance SOC operations and incident response capabilities.

11. Impact Analysis

The implementation of the SOC monitoring environment significantly improved visibility into security events generated across authentication, endpoint, and network sources. By centralizing log collection and analysis within Splunk Enterprise, the project demonstrated how security analysts can efficiently detect suspicious activities, investigate incidents, and correlate related events to gain a comprehensive understanding of system activity.

1. Authentication Monitoring Impact

Authentication monitoring provided visibility into user login activities and enabled the detection of failed authentication attempts.

Impact:

- Improved visibility into authentication events across the monitored environment.
- Enabled early detection of unauthorized access attempts and credential misuse.
- Assisted in identifying potential brute-force and password-guessing activities.
- Enhanced the organization's ability to monitor account security.

2. Endpoint Monitoring Impact

Endpoint monitoring using Sysmon provided detailed information about process executions and system activities occurring on the Windows endpoint.

Impact:

- Enabled detection of reconnaissance and information-gathering activities.
- Improved visibility into process execution events.
- Assisted in identifying potentially suspicious commands and user actions.
- Enhanced endpoint security monitoring and investigation capabilities.

3. Network Monitoring Impact

Network monitoring provided visibility into connections initiated by processes running on the endpoint.

Impact:

- Enabled detection of PowerShell-based network activity.
- Improved monitoring of endpoint communications.
- Assisted in identifying suspicious network connections and reconnaissance attempts.
- Increased awareness of potential lateral movement or network discovery activities.

4. Log Correlation Impact

One of the most significant outcomes of the project was the successful correlation of multiple security events into a single investigation workflow.

Impact:

- Linked authentication, endpoint, and network events into a unified incident timeline.
- Enabled reconstruction of attacker-like behavior based on related activities.
- Improved investigation accuracy by providing additional context around security events.
- Reduced manual effort required during incident analysis.
- Accelerated the investigation and triage process for SOC analysts.

5. Overall Project Impact

The project successfully demonstrated how a Security Operations Center can use centralized monitoring, alert generation, and log correlation to improve threat detection and incident investigation. The implementation provided practical experience with SIEM technologies, security monitoring processes, and SOC workflows while highlighting the importance of visibility, automation, and event correlation in modern cybersecurity operations.

The addition of AI-assisted incident reporting further demonstrated how automation can reduce analyst workload, improve reporting consistency, and enhance overall operational efficiency within a SOC environment.

12. Recommendations

Based on the implementation and analysis performed during this project, several improvements can be made to further enhance the effectiveness of the SOC monitoring environment. These recommendations focus on improving detection accuracy, increasing visibility, reducing analyst workload, and strengthening overall security monitoring capabilities.

1. Alert Tuning

Security alerts should be continuously reviewed and refined to ensure they accurately detect suspicious activities while minimizing unnecessary notifications.

Recommendation:

- Adjust alert thresholds based on organizational requirements.
- Continuously review detection rules and correlation searches.
- Improve alert accuracy through regular testing and validation.

2. False Positive Reduction

Excessive false positives can increase analyst workload and delay incident response.

Recommendation:

- Fine-tune detection logic based on observed system behavior.
- Create exclusions for known legitimate administrative activities.
- Periodically review alert effectiveness and investigation outcomes.

3. Enhanced Endpoint Monitoring

Additional Sysmon configurations can provide greater visibility into endpoint activities and improve threat detection capabilities.

Recommendation:

- Enable monitoring for registry modifications and file creation events.
- Monitor PowerShell command-line activity in greater detail.
- Expand endpoint telemetry collection to support advanced investigations.

4. Multi-Endpoint Monitoring

The current implementation focuses on a single monitored endpoint. Expanding monitoring across multiple systems would provide a more realistic enterprise SOC environment.

Recommendation:

- Deploy monitoring agents on multiple endpoints.
- Centralize log collection from servers, workstations, and network devices.
- Improve visibility across the entire environment.

5. AI-Assisted Incident Reporting

The AI-powered reporting workflow demonstrated the potential of automation in SOC operations.

Recommendation:

- Expand AI capabilities to support incident summarization and prioritization.
- Integrate additional contextual information into generated reports.
- Improve reporting consistency and reduce manual documentation effort.

6. Security Orchestration and Automation (SOAR)

Automation can significantly improve incident response efficiency by reducing manual intervention.

Recommendation:

- Integrate SOAR platforms with SIEM alerts.
- Automate alert enrichment and ticket creation.
- Streamline investigation workflows and response processes.

13. Conclusion

This project successfully demonstrated the implementation of a Security Operations Center (SOC) monitoring environment using Splunk Enterprise and Sysmon. Authentication, endpoint, and network security logs were collected, analyzed, and monitored to identify suspicious activities within a controlled laboratory environment. The project provided practical experience in security monitoring, alert generation, event correlation, and incident investigation, reflecting the responsibilities performed by SOC analysts in real-world environments.

Three security alerts were successfully developed and validated, including Failed Login Detection, System Information Enumeration Detection, and PowerShell Network Reconnaissance Detection. These alerts enabled the identification of suspicious activities across different stages of the simulated attack lifecycle. Through log correlation, multiple security events were linked together to reconstruct attacker-like behavior and create a structured investigation timeline.

The implementation of dashboards provided centralized visibility into authentication events, endpoint activities, network communications, and generated alerts, improving monitoring efficiency and supporting security investigations. In addition, an AI-powered incident reporting workflow was developed as a future enhancement using n8n and Google Gemini to automate alert correlation, incident analysis, and report generation.

Final Outcomes

- Successfully implemented a SOC monitoring environment using Splunk Enterprise and Sysmon.
- Generated and investigated authentication, endpoint, and network security alerts.
- Correlated multiple security events to reconstruct attacker-like behavior.
- Demonstrated the SOC investigation and incident triage workflow.
- Developed a centralized security monitoring dashboard.
- Implemented an automated email notification system for alert delivery.
- Designed an AI-assisted incident reporting workflow using n8n and Google Gemini.
- Demonstrated the complete SOC monitoring lifecycle, including data collection, alert generation, correlation, investigation, dashboard monitoring, and reporting.

The project highlights the importance of centralized log management, security monitoring, event correlation, and automation in modern cybersecurity operations. The integration of artificial intelligence further demonstrates how future SOC environments can improve operational efficiency, reduce analyst workload, and accelerate incident response processes.

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